OpenClaw Servo Control Tutorial

> Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-OpenClaw Prompts], [05-AI Voice Interaction Configuration], and [06-Three Conversation Scheme Configuration and Testing] in the `Preparation for Use` directory. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This document assumes the basic environment is already prepared and only expands on servo-related content.

1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capabilities through servo peripherals. After completing this tutorial, readers will be able to:

- Understand the basic idea of OpenClaw controlling servos
- Interact with OpenClaw via three methods: WhatsApp, TUI, and voice code
- Complete four servo cases: single servo control, servo dancing, yes/no Q&A, simulated ping pong, simulated pointer

2. Hardware Preparation

- OpenClaw main board
- Servo `[1 unit / 4 units]`
- `[Optional] Microphone module`
- Wiring diagram

3. Software Preparation

Code entry points used in the tutorial

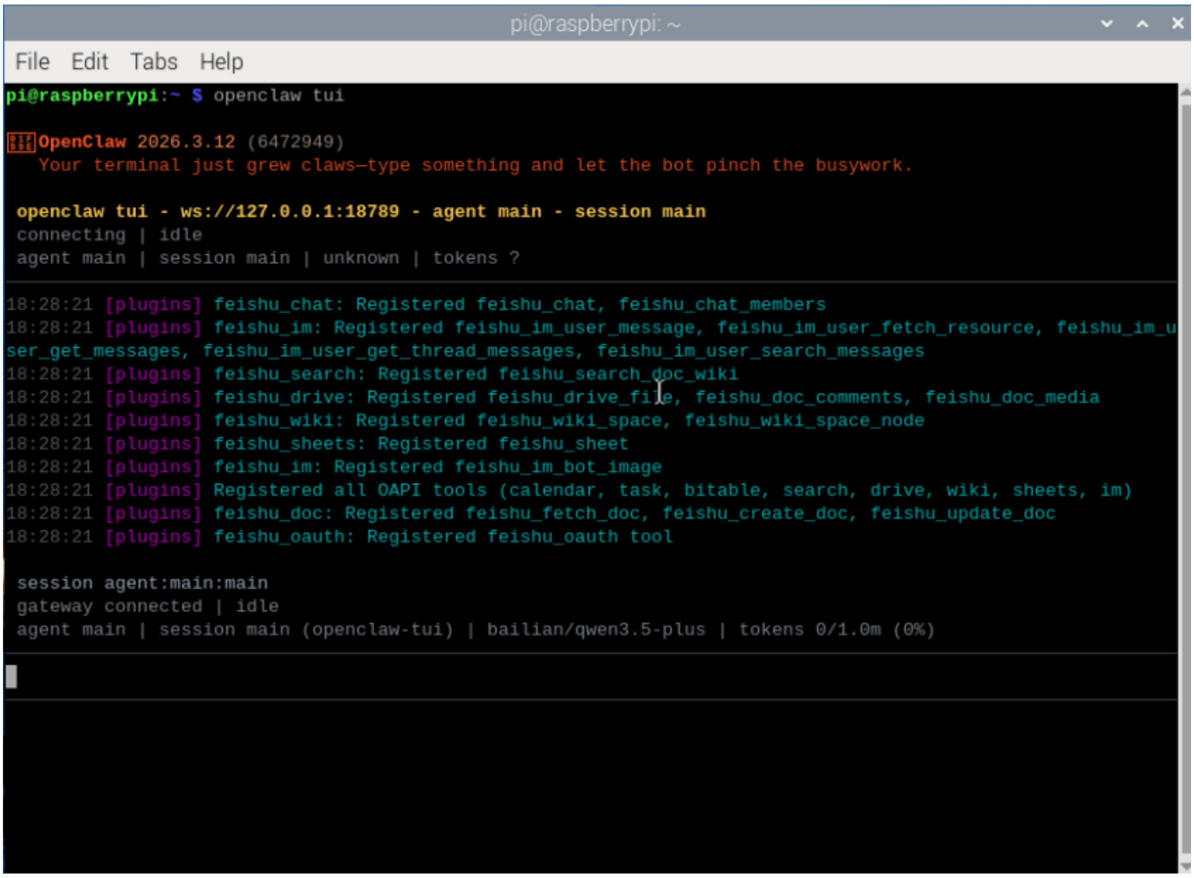
- Low-level driver code: /home/sunrise/hardware_hal_demo/
- Python driver library: /home/sunrise/openclaw-fw/python/
- Voice entry point: `/home/sunrise/voice_to_openclaw/src/main.py`
- Terminal text entry point: `/home/sunrise/voice_to_openclaw/src/text_main.py`

4. Testing the Servo

Before starting this tutorial, please complete a basic `openclaw tui` conversation self-check first; if you haven't done so, see [06-Three Conversation Scheme Configuration and Testing]. After confirming that OpenClaw can respond normally, proceed to the servo-specific test.

Enter the following command in the terminal to enter TUI:

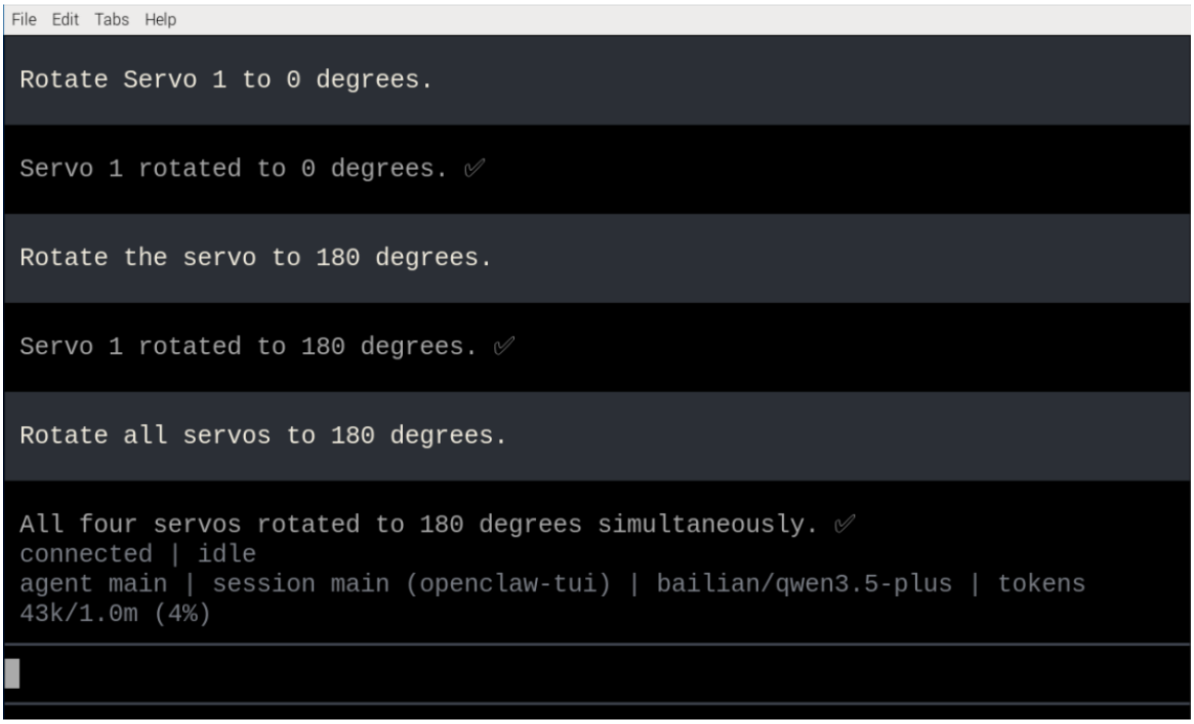
```
openclaw tui
```



After ensuring the hardware is connected, you can test servo control by entering the following content in the terminal:

- Turn servo 1 to 0 degrees
- Turn the servo to 90 degrees
- Turn the servo to 180 degrees

Note: You can specify which servo to control individually.



If you can see the servo driving normally, it means the following points are normal.

- OpenClaw can receive instructions
- Servo power supply is normal
- Servo angle control range is correct
- Clear feedback is visible after instruction execution

5. Three Schemes for Conversing with OpenClaw

Scheme	Suitable Scenarios	Advantages	Suggested Positioning
WhatsApp	Remote control, classroom demonstration	Easy access, suitable for multi-user experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	Get the control pipeline running first
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration scheme

If you have selected the AI voice module + speaker, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

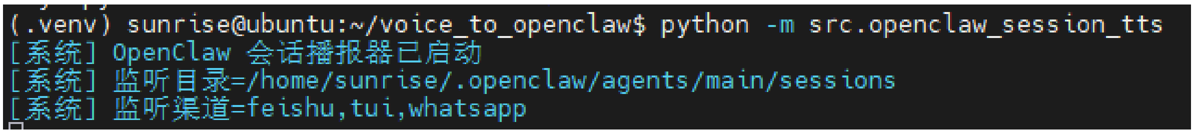
- Modify whether to enable broadcast parameter

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether to broadcast. After modifying and saving, run the following program in the terminal.
```

- Enter in terminal:

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



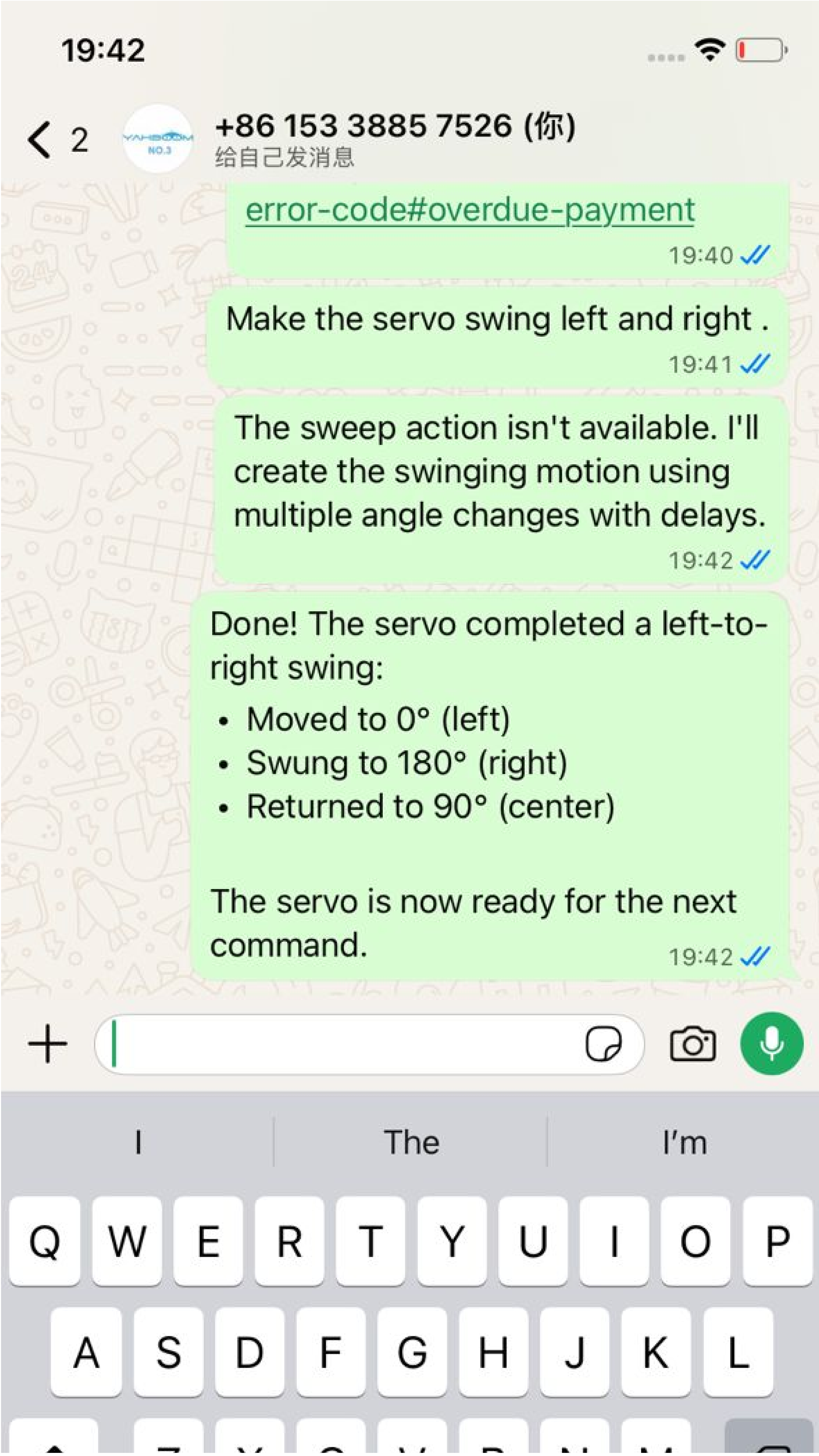
```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Scheme

For WhatsApp access, QR code creation, and gateway restart steps, please refer directly to [06-Three Conversation Scheme Configuration and Testing]. This section only retains conversation examples suitable for the servo tutorial.

You can converse directly, for example:

- Turn servo 1 to 0 degrees
- Turn the servo to 90 degrees
- Make the servo swing left and right once


19:42
+86 153 3885 7526 (你)
给自己发消息
error-code#overdue-payment
19:40
Make the servo swing left and right .
19:41
The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.
19:42
Done! The servo completed a left-to-right swing:
• Moved to 0° (left)
• Swung to 180° (right)
• Returned to 90° (center)
The servo is now ready for the next command.
19:42
I
The
I'm

5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, you can continue the conversation by entering the following command in the terminal:

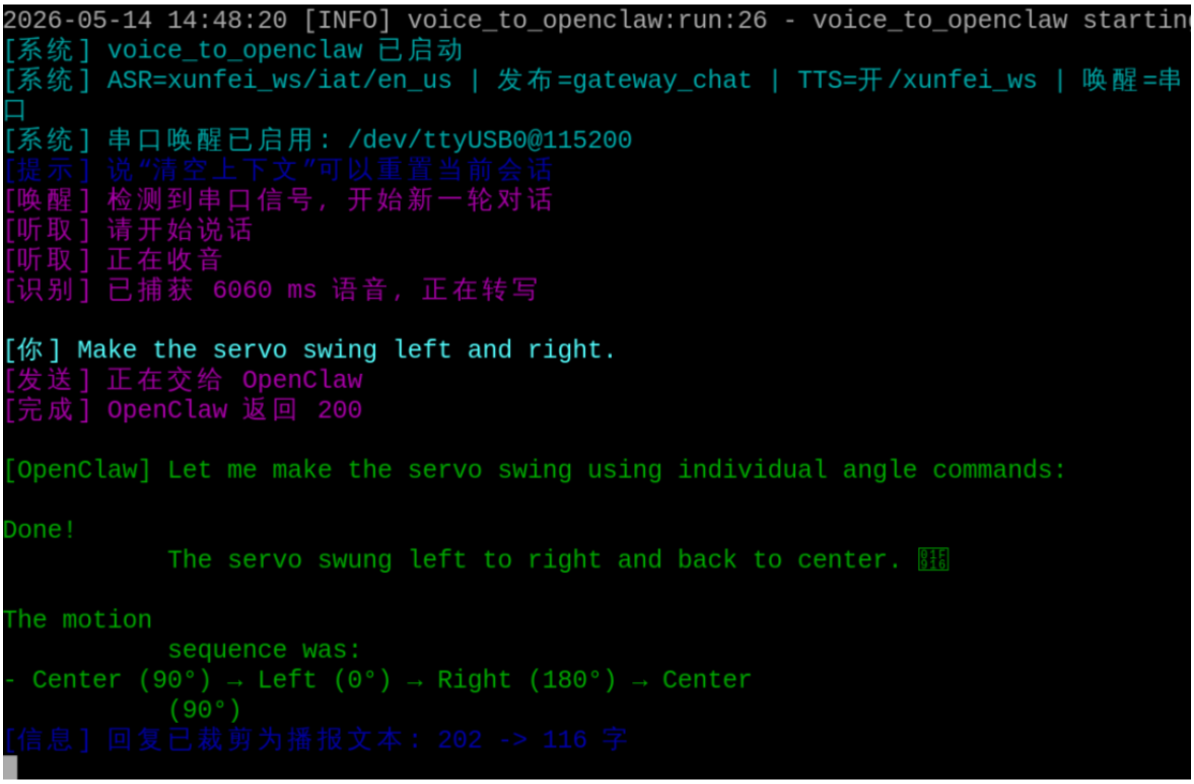
```
openclaw tui
```

5.3 Voice Code Scheme

```
Requires selection of AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. Here it is more suitable to directly demonstrate servo-type oral instructions, for example:

- Turn the servo to 90 degrees
- Make the servo nod
- Make the servo sway left and right



```
2026-05-14 14:48:20 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw startin
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串
口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6060 ms 语音，正在转写

[你] Make the servo swing left and right.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me make the servo swing using individual angle commands:

Done!
           The servo swung left to right and back to center.

The motion
           sequence was:
- Center (90°) → Left (0°) → Right (180°) → Center
           (90°)
[信息] 回复已裁剪为播报文本: 202 -> 116 字
```

6. Comprehensive Cases

Note: The following conversations are conducted via openclaw tui (you can choose your own conversation method).

6.1 Case 1: Servo Dancing

Experiment Objective

Make the servo move continuously according to a set of preset angles, forming a rhythmic effect similar to "dancing."

Effect Description

The servo can switch between multiple angles, such as swinging left, returning to center, swinging right, shaking quickly, and then returning to the initial position. Although there is only one servo, a clear "sense of dance" can still be expressed through changes in movement rhythm.

Example Instructions

- Make the servo dance for 10 seconds
- Make the servo sway left and right with a faster rhythm
- Arrange a three-beat dance step for the servo

6.2 Case 2: Yes/No Q&A

Experiment Objective

Let the servo express "yes" or "no" with actions, for example, nodding for "yes" and shaking head for "no."

Effect Description

When OpenClaw judges the answer as affirmative, the servo performs a nodding action; when the answer is negative, the servo performs a head-shaking action. This case is very suitable for highlighting the combination of "natural language understanding + action expression."

Example Instructions

- If the answer is "yes", make the servo nod
- If the answer is "no", make the servo shake its head
- Help me judge this question and answer with servo actions

6.3 Case 3: Simulated Ping Pong

Experiment Objective

Make the servo perform continuous left-and-right往返 movements, simulating the feeling of a ping pong ball bouncing back and forth between two sides of a table.

Effect Description

The servo swings uniformly or variably between two boundary angles on the left and right. If accompanied by voice description or screen prompts, the audience will easily understand that this is a simulation of "ping pong moving back and forth."

Example Instructions

- Make the servo simulate ping pong moving back and forth
- Swing left and right 20 times, faster speed
- Start from the left side and do a round-trip movement demonstration

6.4 Case 4: Simulated Pointer

Experiment Objective

Use the servo as a rotatable pointer to indicate values, directions, or states.

Effect Description

This case can very intuitively demonstrate the mapping relationship of "abstract value -> actual angle." For example, mapping values from 0 to 100 to angles from 0 to 180 degrees, the servo can point to the target position like a dashboard pointer.

Example Instructions

- Turn the pointer to 30%
- Make the servo point to the 90-degree position
- Simulate a dashboard, current value is 75